

CESAR BATREZ-DELATORRE

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EDUCATION

University Of Chicago

Masters of Science in Computer Science, Advanced Scholar — High Performance Computing focus

Expected May 2028

Syracuse University

Bachelor of Science in Computer Science

Expected May 2027

GPA: 3.8/4.0

Honors/Awards: Catalyst Scholar, Success Scholar, Dean's List, SOURCE Bridge Grant Recipient

SKILLS

Languages: C, C++, Python, Java, JavaScript, Haskell, MATLAB

Libraries/Frameworks: NetworkX, NumPy, SciPy, Git

TECHNICAL EXPERIENCE

Optimal Market Making under Inventory Risk

August 2026 - Present

Python, C++, pybind11, NumPy, pandas, CMake

- Implemented the Avellaneda-Stoikov optimal market-making model, framing quote placement as a stochastic optimal control problem that balances spread capture against inventory risk via an inventory-adjusted reservation price
- Developed a market-making simulator with a performance-critical C++ core exposed to a Python research layer via pybind11, mirroring the compiled-core / research-layer split of production quant systems

Syracuse University

January 2026 - May 2026

Undergraduate Researcher

Syracuse, NY

- Developed scalable matrix approximation algorithms by designing gradient-based methods and tridiagonal Toeplitz matrices, optimizing computational efficiency and processing speed while preserving critical spectral properties
- Engineered query-efficient optimization techniques, building stochastic gradient descent (SGD) algorithms with randomized matrix-vector products and trace estimators to minimize query complexity on large-scale datasets

Syracuse University: SOURCE

January 2025 - April 2025

Undergraduate Researcher

Syracuse, NY

- Engineered a graph-based transit routing model in Python utilizing NetworkX to map a 36-stop network spanning 40+ unique routes, optimizing spatial data structures for efficient traversal
- Implemented Dijkstra's shortest-path algorithm to design custom route-optimization software, identifying critical transit bottlenecks and reducing path latencies by up to 27

WORK EXPERIENCE

Syracuse University

August 2026 - Present

Undergraduate Teaching Assistant - CIS 375: Discrete Mathematics

Syracuse, NY

- Led recitation sections and technical consultations for 50 students, accelerating comprehension of set theory, mathematical induction, graph theory, and symbolic logic
- Evaluated 50+ weekly examinations and algorithmic proofs, diagnosing systemic logical errors and enforcing rigorous mathematical reasoning standards

Syracuse University

August 2025 - Present

AEW Facilitator - CIS 151: Intro to Computer Science

Syracuse, NY

- Facilitated weekly technical workshops to drill 15 students on core computing fundamentals, including Object-Oriented Programming (OOP), control flow, and recursion
- Mentored student groups in defensive programming, structural testing, and collaborative debugging strategies to optimize runtime performance

ClosetFullOfCoco

February 2026 - August 2026

Generative 3D / Web Developer

San Francisco, CA

- Led 3D visual pipeline production for a luxury e-commerce startup, leveraging Blender API and Python to programmatically generate 60+ high-resolution product renders and dynamic media asset pipelines
- Collaborated directly with executive founders to scale platform storytelling, managing digital assets for a 300+ product catalog across web architecture and marketing channels

Lavner Education

May 2025 - August 2025

Technology Instructor

Chicago, IL

- Instructed 10+ students weekly in intermediate programming tracks, detailing algorithm design and software paradigms across Python and JavaScript
- Designed 20+ interactive coding practicums and algorithmic challenges to reinforce problem-solving, structural design, and code optimization practices